UG (NEP) 2nd SEMESTER
COMPUTER APPLICATIONS (MAJOR) — CAP222J

Session: July 2023 No. of Questions: 09
Time Allowed: 2:00 Hours Maximum Marks: 100 Minimum Marks: 40

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. What is source code and object code?
2. What is a recursive function?
3. What is a variable? How is a variable declared in C language?
4. What is a unary operator? Give an example.
5. Differentiate between break and continue statements with an example.
6. What is a pointer? How is a pointer initialized?
7. What is a string?
8. What is dynamic memory allocation?

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. What is a flowchart? What are the different symbols used in a flowchart? Draw a flowchart for finding whether a number is even or odd.

OR

What is an algorithm? What are its characteristics? Write an algorithm to find the sum of numbers from 1 to N.

3. What is an operator? What are the arithmetic operators available in C language? Give an example of each.

OR

Write a program in C to find the greatest of three numbers.

4. Explain loop control statements of C language with examples.

OR

Write a program in C to print the sum of digits of a number.

5. What is an array? Write a C program to sort elements of an array in ascending order.

OR

Write a C program to implement the following library functions: (a) strcpy (b) strcat (c) strcmp (d) strlen.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. (a) What is a programming language? Discuss the importance of programming languages.
   (b) What is sequence control? Explain various categories of sequence control.
7. What is a datatype? What are the different datatypes available in C language? Explain with examples.
8. What is a function? Explain function call, function definition and function prototype. Differentiate between call by value and call by reference with an example.
9. What is a structure? How is it different from a union? Write a program in C to sort an array of structures.